VANISHING OF H^1 TWISTED DUALIZING MODULE FOR GLOBALLY GENERATED LINE BUNDLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: two steps are implicit in the source and their mechanisms are recorded here: (a) the existence of s with st nonzero is the generator argument - t is a map L -> O_X and vanishing on a generating family of global sections would force t = 0; (b) a nowhere-vanishing global section of an invertible sheaf trivializes it, so L cong O_X; the duality input is the in-source Riemann-Roch section (navigation stub).

Editorial note. The label lemma-vanishing-twist names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Serre duality on the Cohen-Macaulay curve. Same unit, Situation situation-Cohen-Macaulay-curve and the duality of Section section-Riemann-Roch. The proof reduces H^1(X, omega_X tensor L) = 0 to H^0(X, L^{tensor -1}) = 0 by the duality discussed in the Riemann-Roch section.

Source reference. Original section 0B5B: sources/stacks-curves/pinned-original-source.tex, source lines 1109–1109. Prerequisite pointer invoked at proof line(s) 1362; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0B5D: sources/stacks-curves/pinned-original-source.tex, source lines 1307–1307. Prerequisite pointer invoked at proof line(s) 1355; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 0B5D. Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module which is globally generated and not isomorphic to $\mathcal{O}_X$. Then $H^1(X, \omega_X \otimes \mathcal{L}) = 0$.*

Proof. By duality as discussed in Section 0B5B we have to show that $H^0(X, \mathcal{L}^{\otimes -1}) = 0$. If not, then we can choose a global section t of $\mathcal{L}^{\otimes -1}$ and a global section s of $\mathcal{L}$ such that $st \neq 0$. However, then st is a constant multiple of 1, by our assumption that $H^0(X, \mathcal{O}_X) = k$. It follows that $\mathcal{L} \cong \mathcal{O}_X$, which is a contradiction. □

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